

Transformations of $y=x$



- 1
 - a Yes
 - b No
- 2
 - a They become very large in the negative direction.
 - b They become very large in the positive direction.
- 3
 - a Increasing
 - b Equally steep
- 4
 - a No
 - b No
 - c No
 - d Yes
- 5
 - a No
 - b No
 - c No
 - d Yes
- 6
 - a The constant is the same.
 - b All of the graphs cross the y -axis at the same point.
 - c Equations of the form $y = mx + b$ that have the same value of b generate graphs that cross the y -axis at the same point.
- 7
 - a $y = 1$
 - b b
- 8
 - a
 - i $x = 0$
 - ii $y = 0$
 - b
 - i $x = 3$
 - ii $y = 4$
 - c
 - i There is no x -intercept.
 - ii $y = -1$
- 9
 - a No
 - b No
 - c Yes
 - d Yes
 - e No
 - f Yes

10 Positive



11

a Yes

b Yes

c No

d No

e Yes

12

a $y = 5$

b $y = 0$

c $y = 10$

d $y = 1$

13

a Yes

b No

c Yes

d Yes

e Yes

14

a $(4, 0)$

b $(0, -4)$

c $(0, -10)$

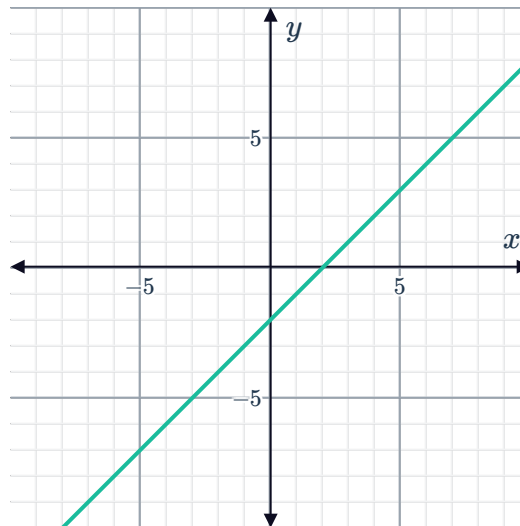
15

a $(4, 0)$

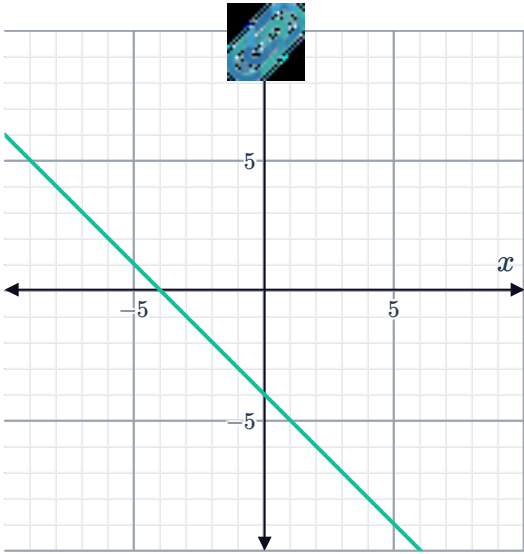
b $(0, 4)$

c $(0, 14)$

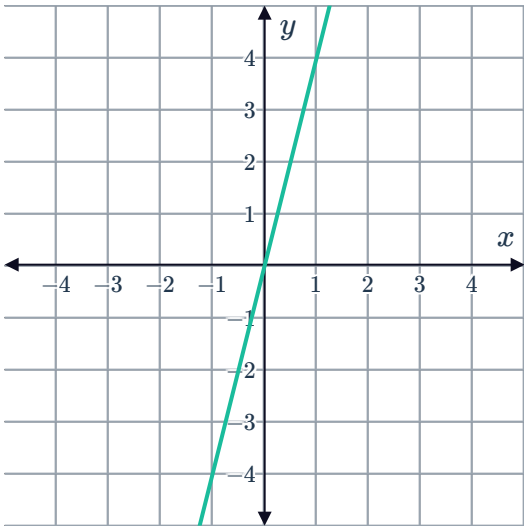
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17



18



19

- a

Yes
- b

Yes
- c

No
- d

No
- e

No

20

- a

Yes
- b

No
- c

Yes
- d

No
- e

No
- f

No

21 (1,3)

22

- a

Increasing
- b

Flatter
- c



23

a

- i Yes ii No iii No iv Yes
v Yes

b

- i No ii Yes iii Yes

c $x = 8$

24

a Decreasing

b Steeper

c

- i Yes ii Yes iii No iv No
v No

25

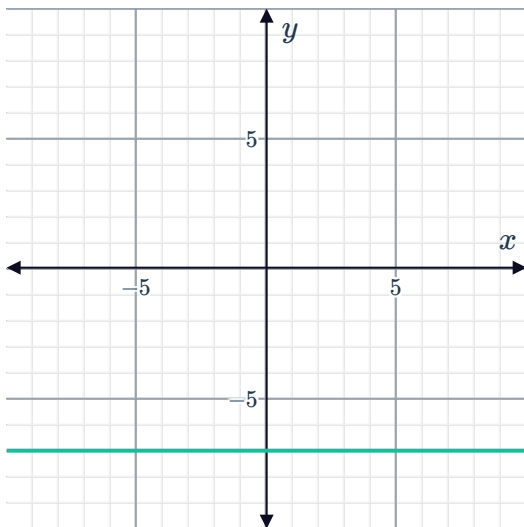
- a Statement 2: The set of points lie on a horizontal line.
b Statement 2: The set of points lie on a horizontal line.
c Statement 1: The set of points lie on a vertical line.
d Statement 3: The set of points lie on an increasing line.
e Statement 1: The set of points lie on a vertical line.
f Statement 4: The set of points lie on a decreasing line.

26

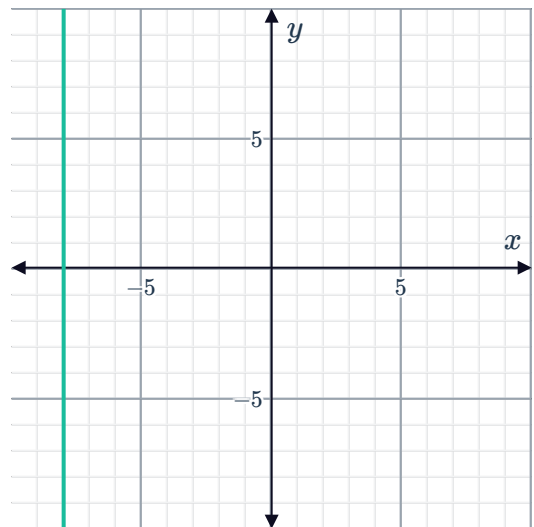
- a A horizontal line b A vertical line

27

a



b



28

a $x = -8$

b None

29

a $y = -4$

b None

30

a $x = -3$

b $y = 4$

31 $y = -4.5$

32 x -axis

33 y -axis

34

a $y = 13, y = -7$

b $x = 5, x = -9$

35 $y = 0, y = -4$

36

a \$2.50

b A fixed cost such as a toll charge



Additional questions

37

- a The transformation from $f(x)$ to $g(x)$ is a vertical translation.
- b The transformation from $f(x)$ to $g(x)$ is making the line steeper.
- c The transformation from $f(x)$ to $g(x)$ is a reflection across the x -axis.

38 When k is negative, the transformation will involve a reflection across the x -axis.

39

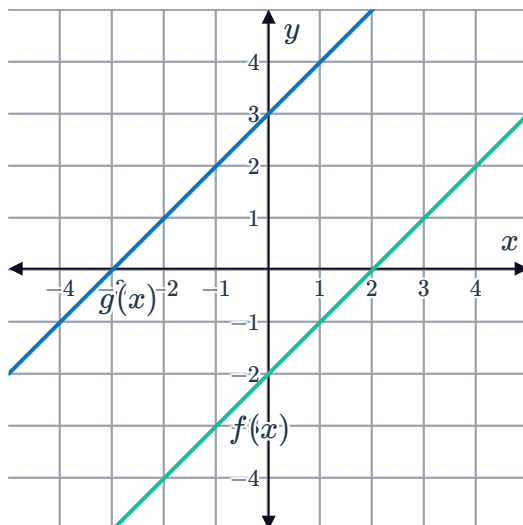
- a Vertical translation
- b Making the line steeper
- c Making the line less steep
- d Making the line less steep
- e Reflection
- f Vertical translation

40

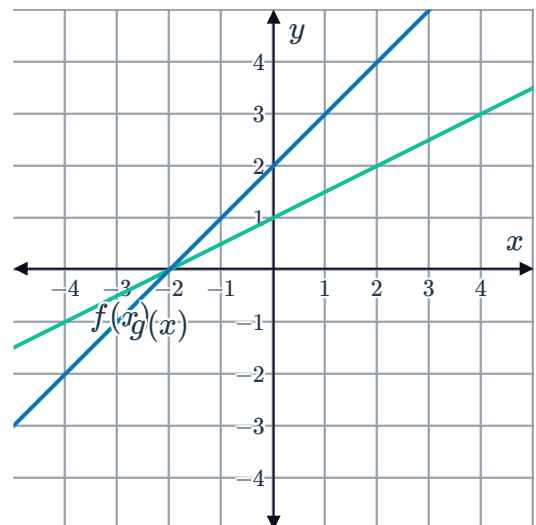
- a Vertical translation
- b Making the line steeper
- c Making the line less steep and reflection across the x -axis
- d Vertical translation

41

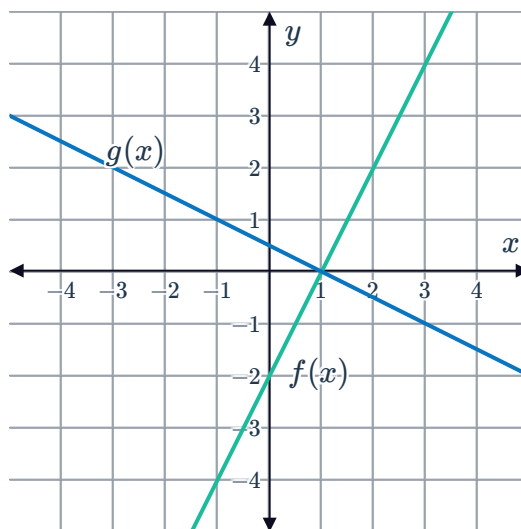
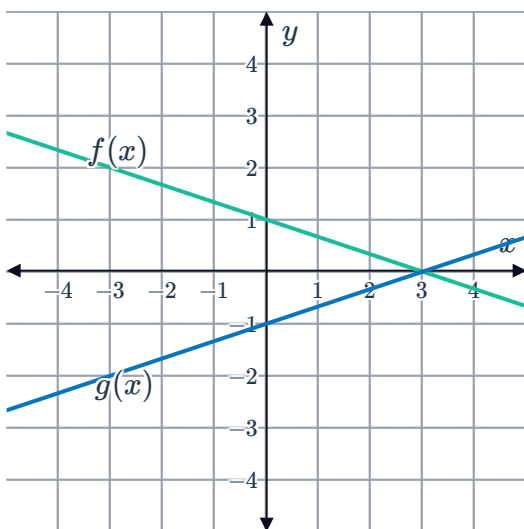
a



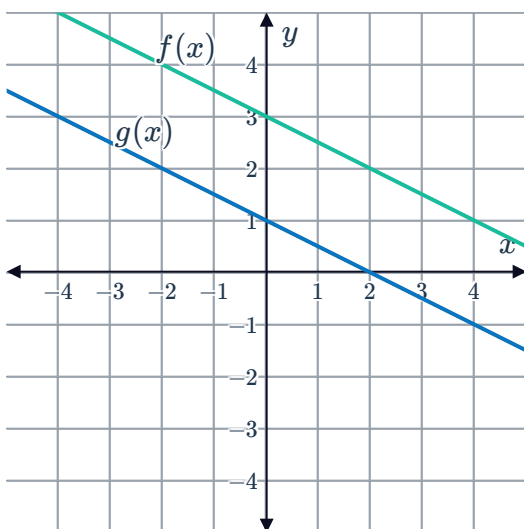
b



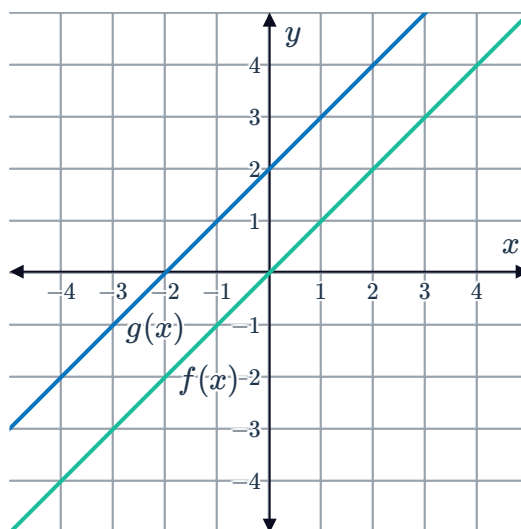
c



e



f



42

- a The graph of $g(x)$ is the vertical translation of 8 units downwards from the graph of $f(x)$.
- b The graph of $g(x)$ is the vertical translation of $\frac{1}{2}$ units upwards from the graph of $f(x)$.
- c The graph of $g(x)$ is steeper than the graph of $f(x)$ by a factor of -3 and a reflection across the x -axis.
- d The graph of $g(x)$ is making the graph of $f(x)$ steeper by a factor of $\frac{1}{4}$. This is equivalent to making the line less steep by a factor of 4.

43

- a Oreste reflected the graph of $f(x)$ across the y -axis instead of the x -axis.
- b If $f(x)$ is a linear function passing through the origin, then a reflection across the y -axis has the same effect as a reflection across the x -axis. If this is the case, Oreste's error would still give the correct answer.

